

# Econ 104 Project

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```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.1.2
```

```
## Warning: package 'tibble' was built under R version 4.1.2
```

```
## Warning: package 'forcats' was built under R version 4.1.2
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr    1.5.1
```

```
## v ggplot2    3.5.1      v tibble     3.2.1
```

```
## v lubridate  1.9.3      v tidyr      1.3.1
```

```
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
```

```
library(MASS)
```

```
## Warning: package 'MASS' was built under R version 4.1.2
```

```
##
```

```
## Attaching package: 'MASS'
```

```
##
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

```
##      select
```

```
library(dplyr)
```

```
library(corrplot)
```

```
## corrplot 0.94 loaded
```

## Section 1

```
“rm(list = ls())”
```

## Data

```
data <- read_csv("data.csv")
```

```
## Rows: 1833 Columns: 10
## -- Column specification -----
## Delimiter: ","
## dbl (9): SP500, Dividend, Earnings, Consumer Price Index, Long Interest Rat...
## date (1): Date
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
sandp <- data[c(1, 2, 6, 8, 9)]
```

```
sandp <- rename(sandp, "Interest" = "Long Interest Rate", "Div" = "Real Dividend", "Earn" = "Real Earnings")
```

```
sandp$SP500 <- as.numeric(sandp$SP500)
sandp$Earn <- as.numeric(sandp$Earn)
sandp$Interest <- as.numeric(sandp$Interest)
sandp$Div <- as.numeric(sandp$Div)
```

```
sandp <- sandp[-c(1830:1833), ]
```

## Variables

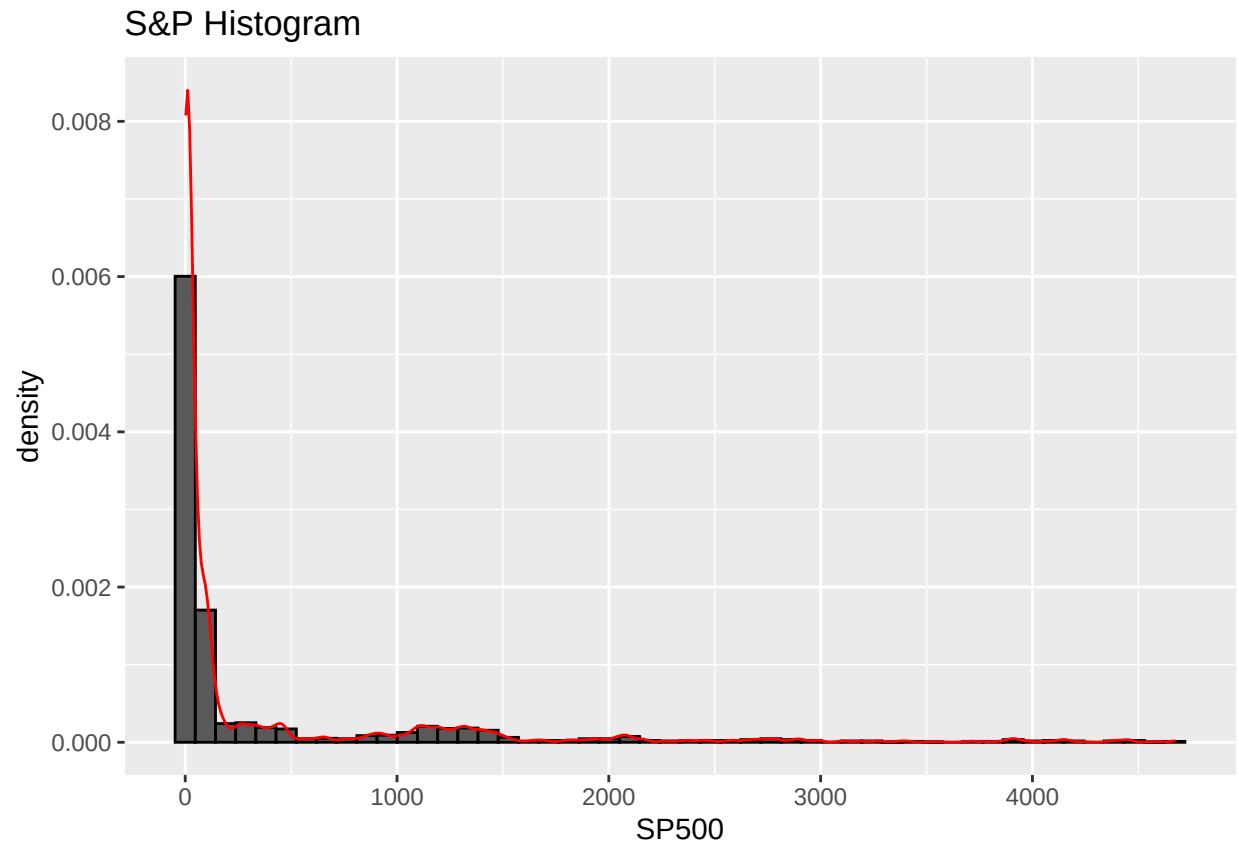
The dataset I will be using for this project is a monthly tracking of the S&P 500 stock price dating back to 1871. The dataset includes the price of the stock on the first of each month, but also includes the inflation rate for said month, the dividend paid by the stock, and the Real Earnings. These four numerical, continuous variables will be the center of the project. I will be using the S&P value and the Real Earnings as my y1t and y2t, and the Dividend payments and Long Interest Rate as the predictors. Because S&P and Earnings share dynamics, our VAR prediction should work well.

## Question 1: Analysis of Variables

### Histograms

```
ggplot(data = sandp, aes(x = SP500)) +
  geom_histogram(aes(y = ..density..), bins = 50, color = "black") +
  geom_density(col = "red") +
  ggtitle("S&P Histogram")
```

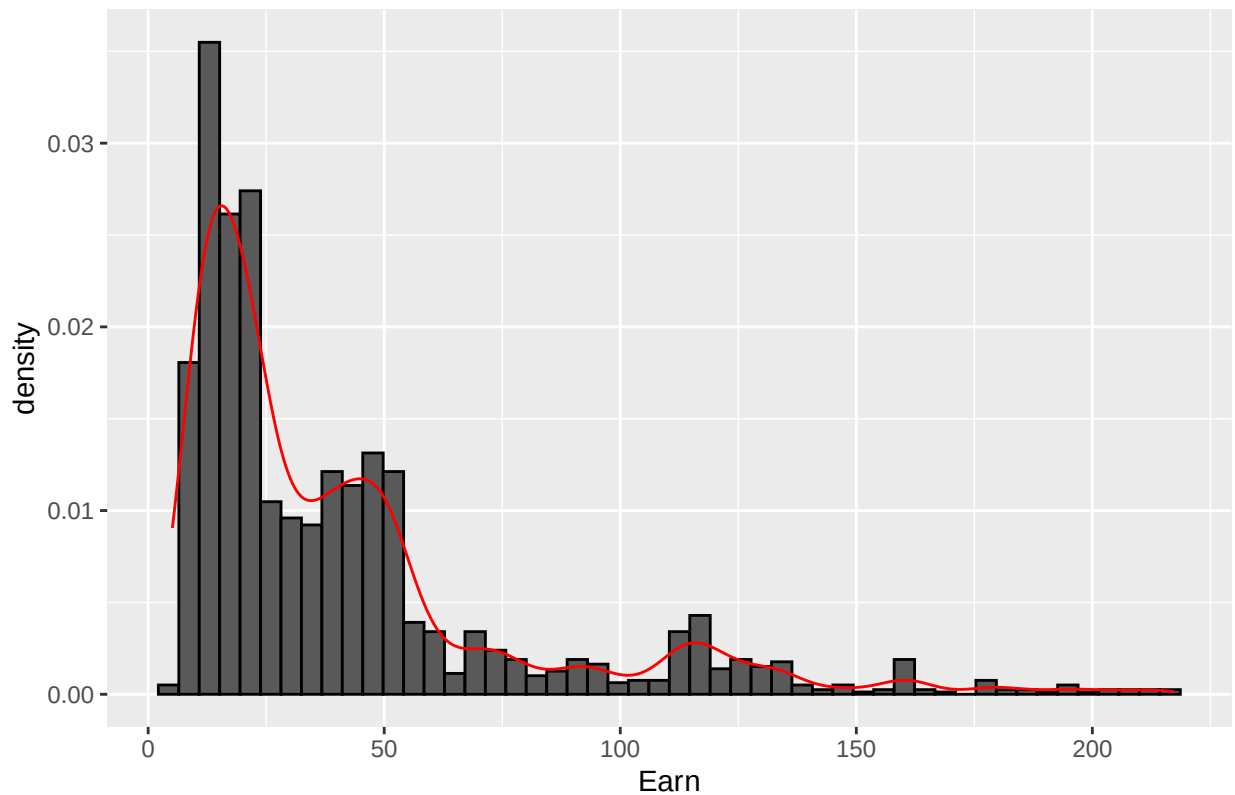
```
## Warning: The dot-dot notation ('..density..') was deprecated in ggplot2 3.4.0.
## i Please use 'after_stat(density)' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



The S&P index is a continually rising figure, so the boxplot is heavily skewed to the right. The index has been around since 1871, and didn't break 100 points until 1979. Now it is at 3000. This exponential growth would be best represented by an exponential curve, which is what the red line most looks like (just reversed).

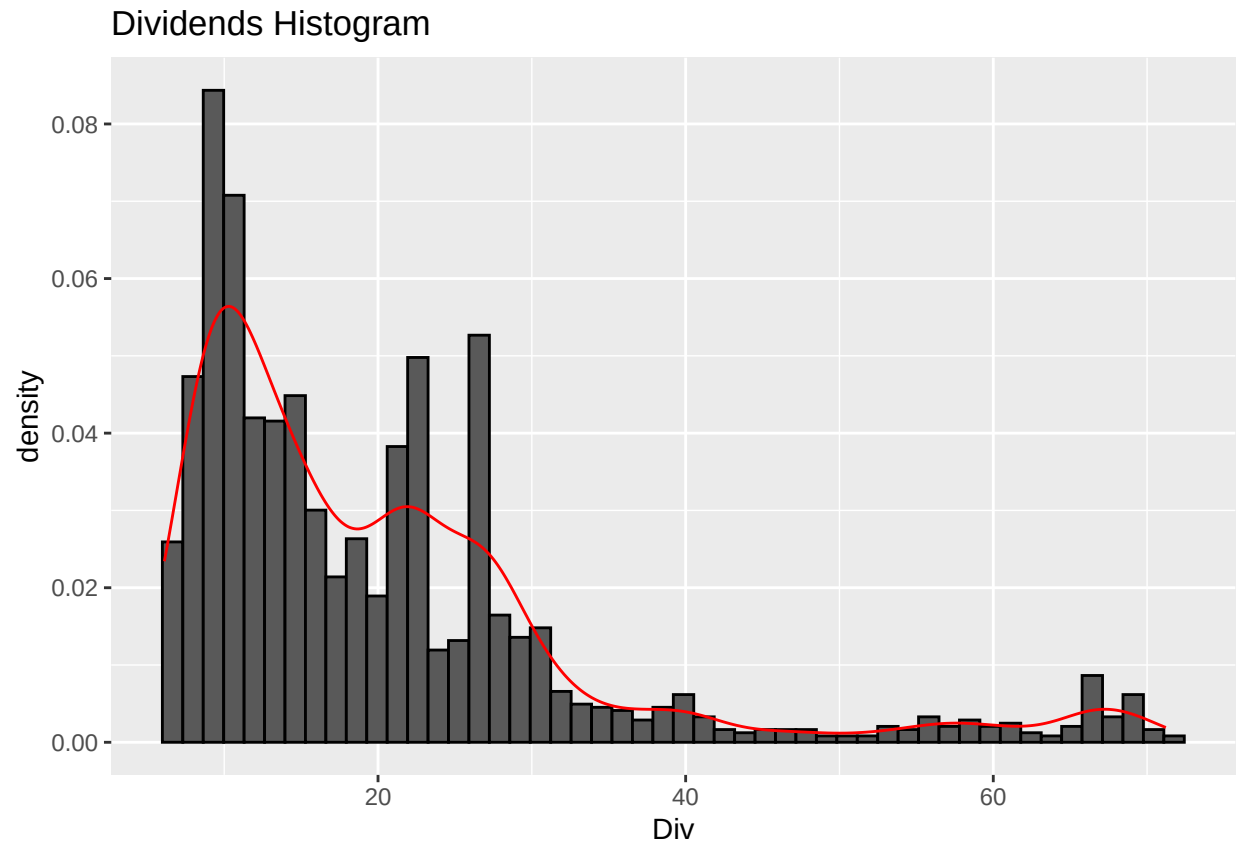
```
ggplot(data = sandp, aes(x = Earn)) +  
  geom_histogram(aes(y = ..density..), bins = 50, color = "black") +  
  geom_density(col = "red") +  
  ggtitle("Earnings Histogram")
```

### Earnings Histogram



The Earnings data is also right skewed, but not as significantly as S&P. This density curve (red line) looks like a multi-tiered log-normal curve. It could also be described as multiple log-normal curves layered on top of each other. This would mean that each earning bracket experiences mirrored but higher earnings, and can definately be used in a model.

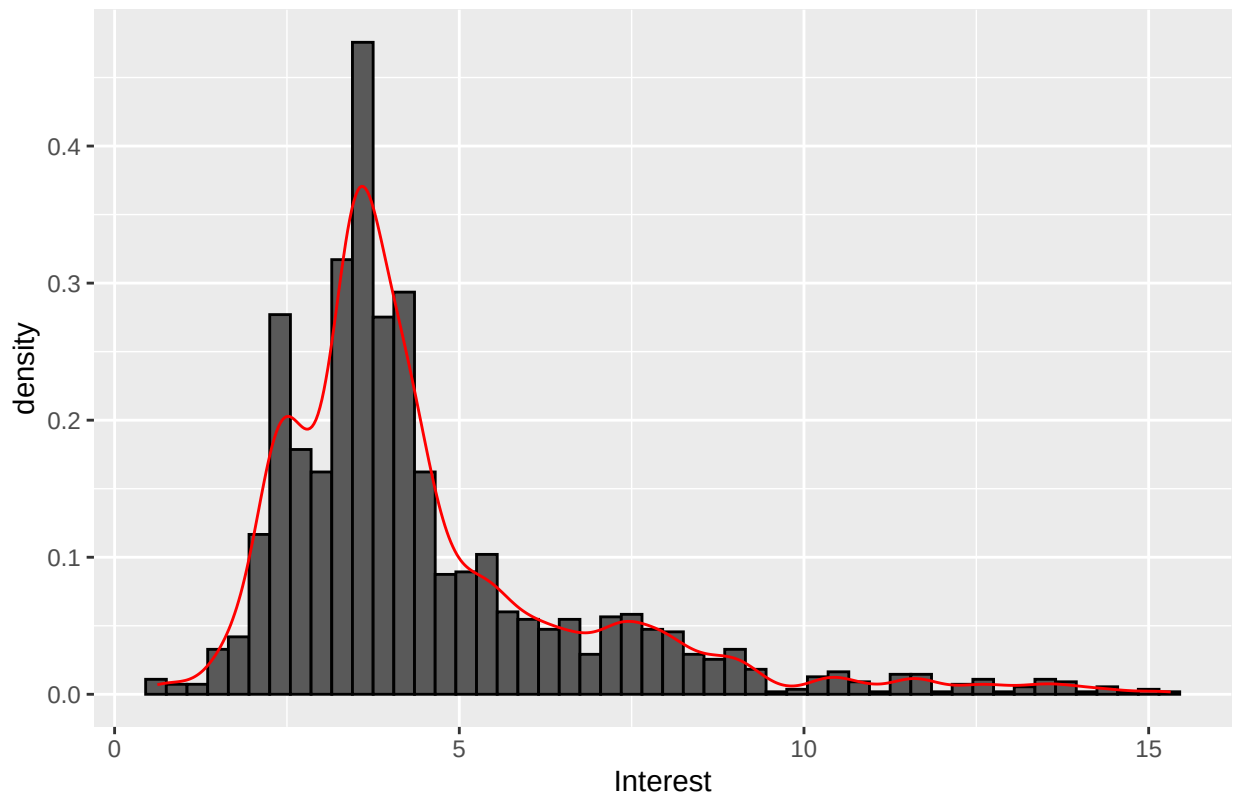
```
ggplot(data = sandp, aes(x = Div)) +  
  geom_histogram(aes(y = ..density..), bins = 50, color = "black") +  
  geom_density(col = "red") +  
  ggtitle("Dividends Histogram")
```



The Dividends graph is very similar to the Earnings graph, which usually means a high correlation between the two. The density curve is doesn't quite reach the peaks (outliers or patterns may be present), but we see the same, right skewed pattern with mini peaks all the way down the line.

```
ggplot(data = sandp, aes(x = Interest)) +  
  geom_histogram(aes(y = ..density..), bins = 50, color = "black") +  
  geom_density(col = "red") +  
  ggtitle("Interest Rate Histogram")
```

## Interest Rate Histogram



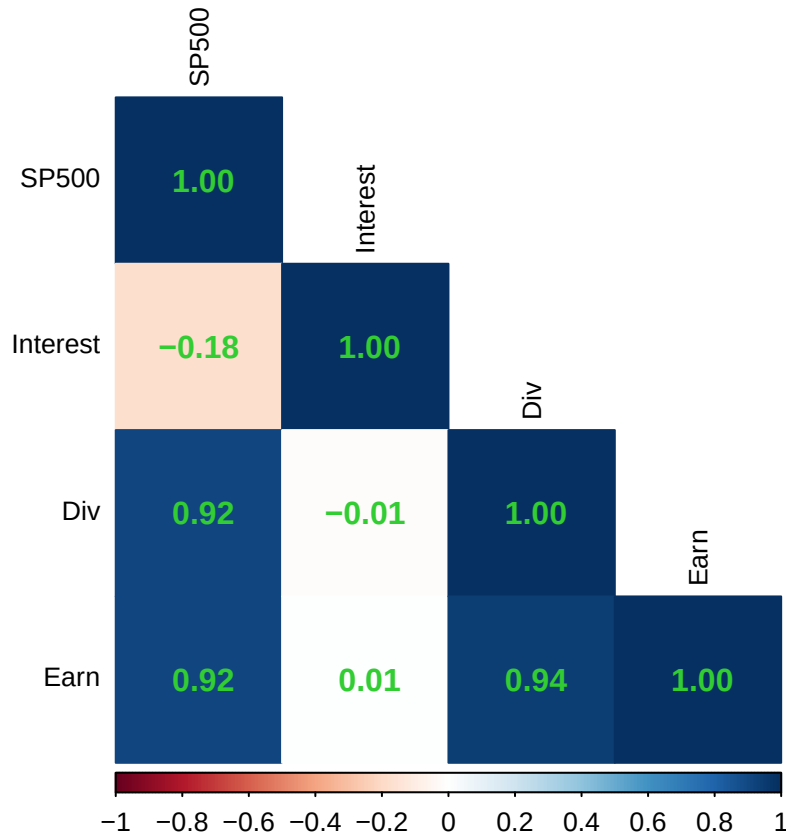
The Interest rate is supposed to be a number that doesn't change too quickly, so more central data is expected, and the histogram supports this. There is still a right tail, but the peak is much more normal this time, and almost all of the mini-peaks that we had seen down the right line in previous figures are now gone. For those reasons, Interest should prove to be a good predictor of SP500 and Earnings.

## Correlation Plot

```
sandp2 <- sandp[, -1]

corr_matrix <- cor(sandp2)

corrplot(corr_matrix, method = "color", type = "lower", addCoef.col = "limegreen", tl.col = "black", tl
```



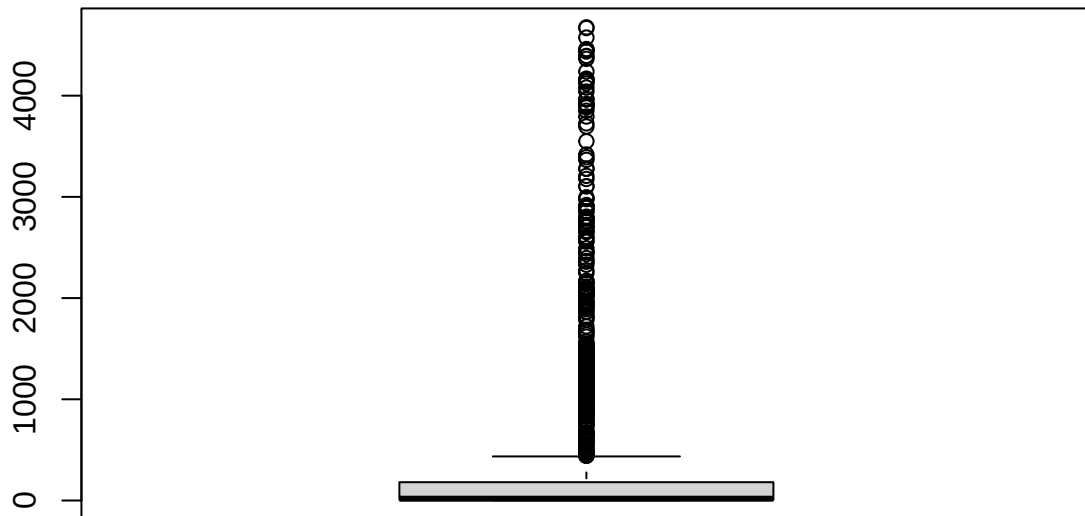
This Correlation Plot, generated with the `corrplot` library, provides a quick way to see how closely variables are correlated with each other. The darker the color, the more significant the correlation. I used a new dataset (`sandp2`) for this problem to eliminate the date variable from the chart.

Each variable is fairly correlated to each other. The least significant correlation is found between the real interest rate and each of the other three variables. It is good to see a high correlation between S&P and Earn, as these are our two y values for the VAR test. We also see a negative correlation between the Interest Rate and the S&P. This is logically correct because when the interest rate is higher, people can make more money leaving their savings in the bank compared to investing, and prices should go down as a result. Interest Rates and Dividend payments have no relationship with each other, but this is because a dividend payment only reflects the stock price and not the rest of the market (same logic for Earn and Interest). Overall, this correlation graph is very promising.

## Boxplot

```
boxplot(sandp$SP500, main = "S&P Boxplot")
```

## S&P Boxplot

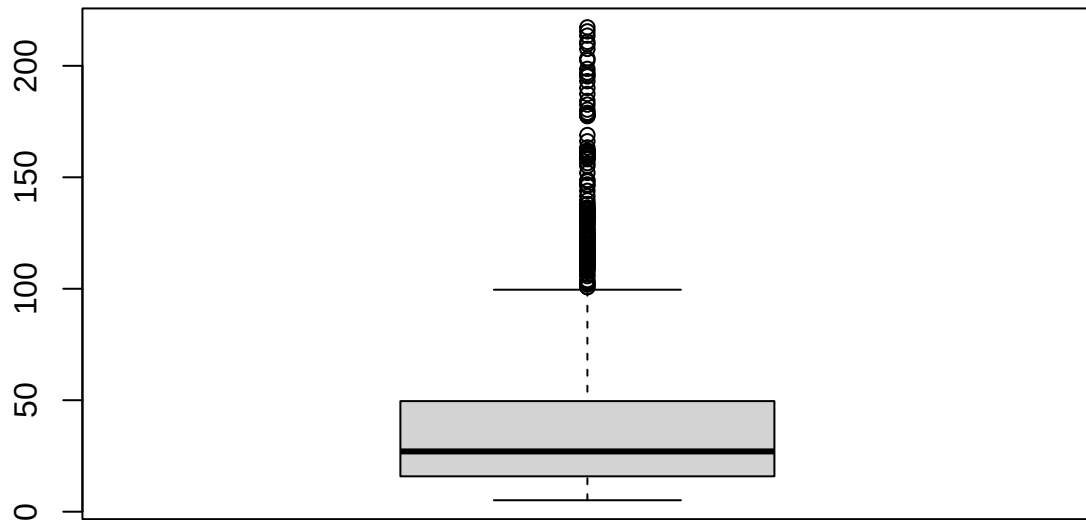


Due to the nature of the market (increasing exponentially), the median is very low compared to current stock prices. Compounding numbers monthly will always lead to a misleading median, so a boxplot is not a good representation for this variable

```
boxplot(sandp$Earn, main = "Earnings Boxplot")
```



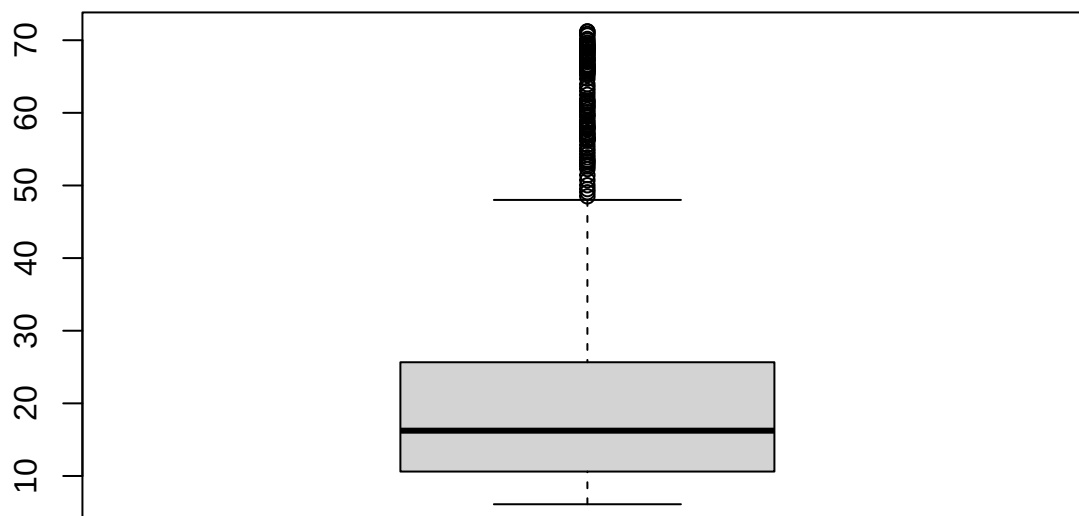
## Earnings Boxplot



Similar to S&P, the Earnings is steadily increasing over time, but less significantly. There are many less outliers here than in the previous boxplot, and the quartiles have a little bit more meaning. The median is still misleading, but this boxplot does show that CPI grows much slower than the S&P.

```
boxplot(sandp$Div, main = "Dividend Boxplot")
```

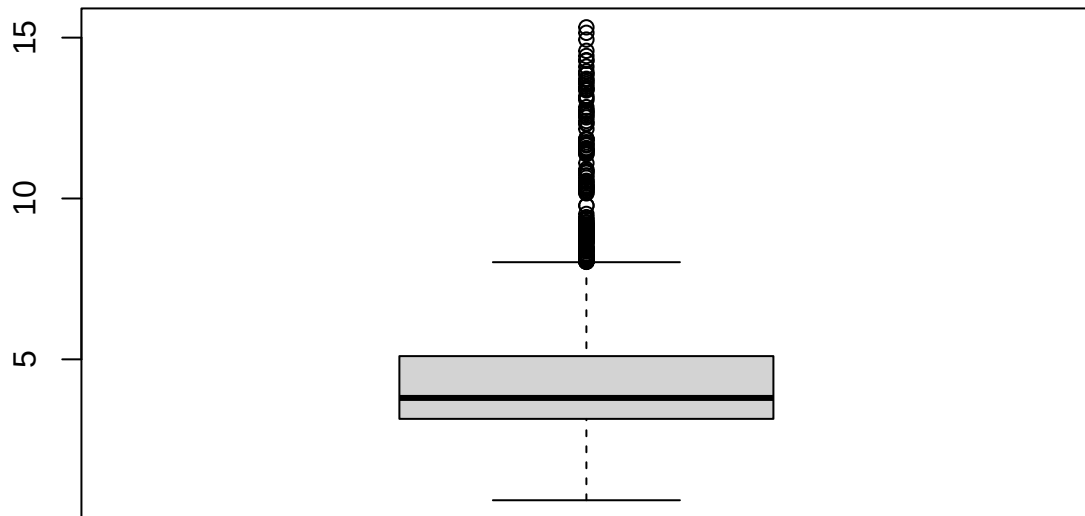
## Dividend Boxplot



This boxplot shows much more than the prior two plots because the dividends don't rely much on the time. The Dividend payment fluctuates more, and has a cyclical nature. The median here is around 12 dollars, with anything over 50 being considered an outlier. Half of the 1833 dividend payments fall between 10 and 25 dollars.

```
boxplot(sandp$Interest, main = "Interest Boxplot")
```

## Interest Boxplot



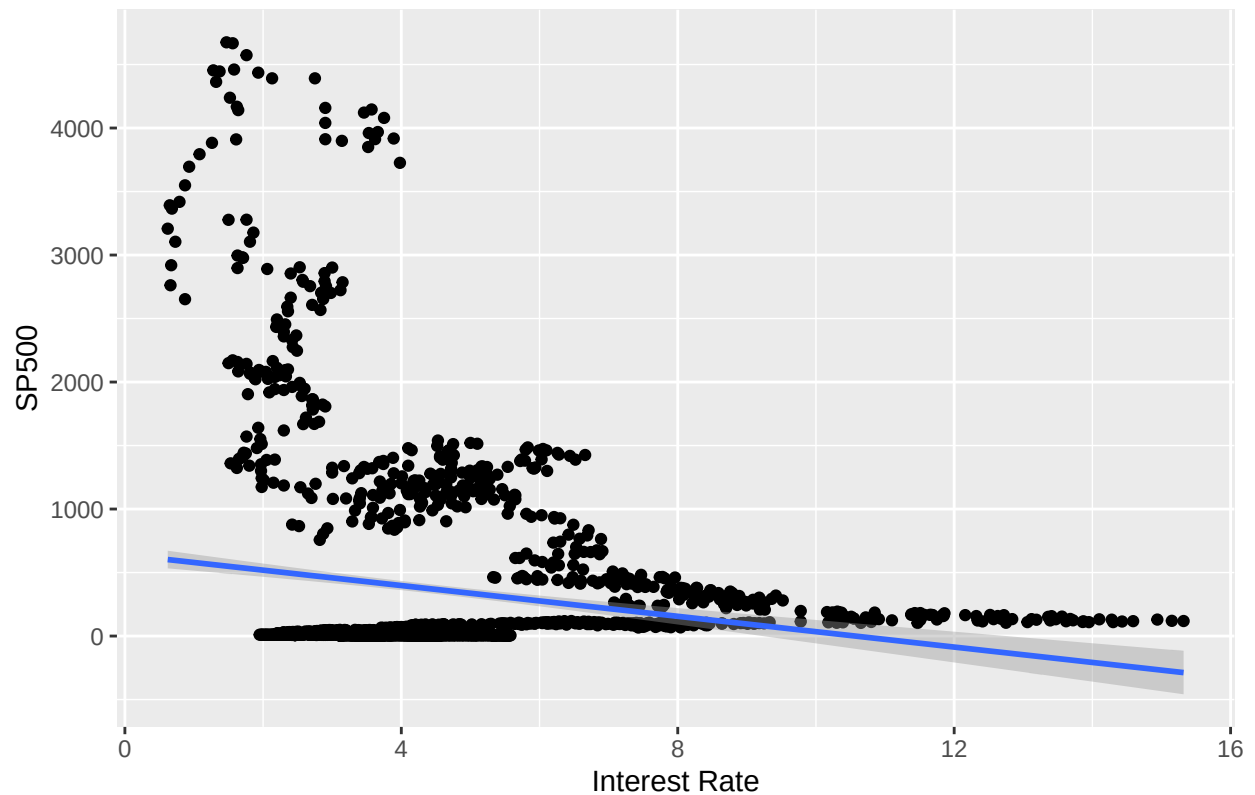
Interest and the S&P are mostly independent. One can be an indication of the other, but the only way one can influence the other is if the numbers affect someones investment decision. Interest rates tend to be constant around 3%, and that can be seen here. A few outliers (anything about 7.5%) push the median to 4%, but half of the data falls between 3 and 5% historically.

## Scatter Plots

```
ggplot(data = sandp, aes(x = Interest, y = SP500)) +  
  geom_point() +  
  geom_smooth(method = "lm") +  
  ggtitle("Interst/SP500 Scatter") +  
  labs(x = "Interest Rate", y = "SP500")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Interst/SP500 Scatter

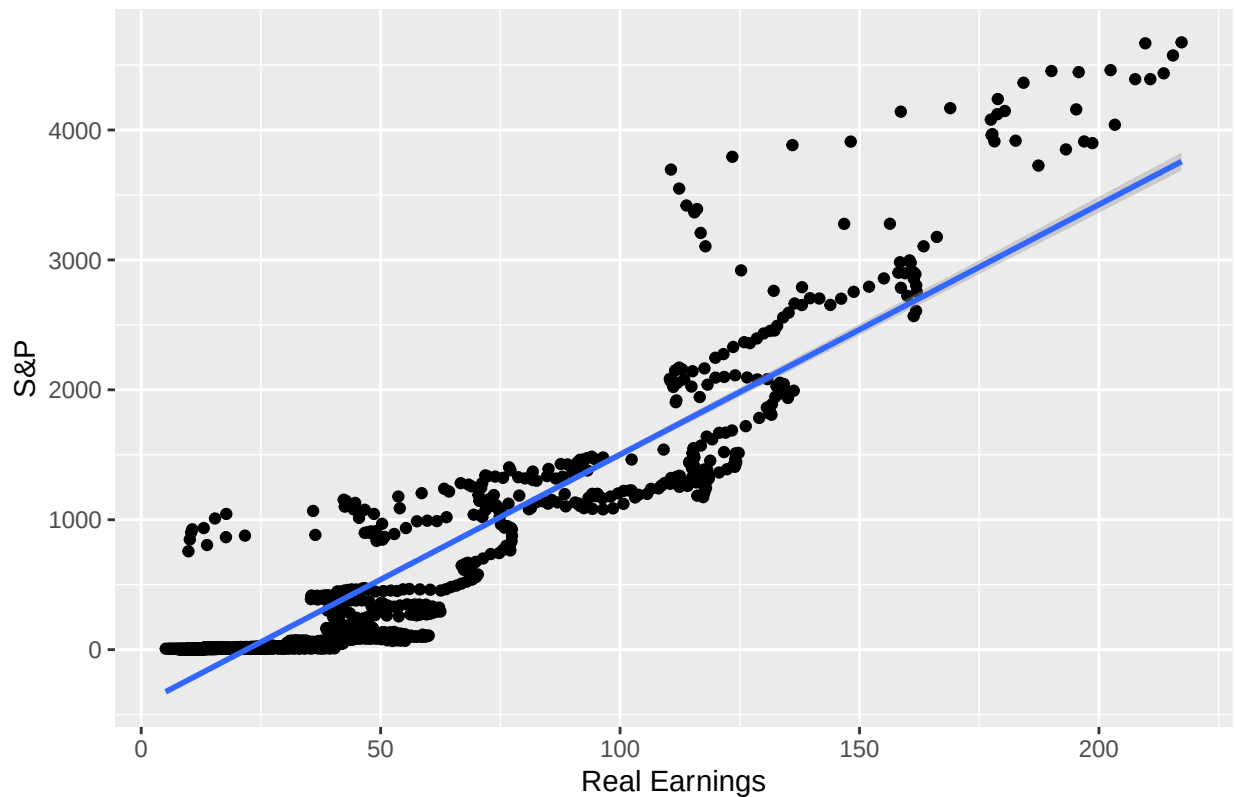


As dictated by our correlation figure, a lower interest rate leads to a higher SP500 figure. The blue line of best fit reflects this negative relationship. The correlation is weak but the pattern is clear.

```
ggplot(data = sandp, aes(x = Earn, y = SP500)) +  
  geom_point() +  
  geom_smooth(method = "lm") +  
  ggtitle("Earnings/S&P Scatter") +  
  labs(x = "Real Earnings", y = "S&P")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

## Earnings/S&P Scatter

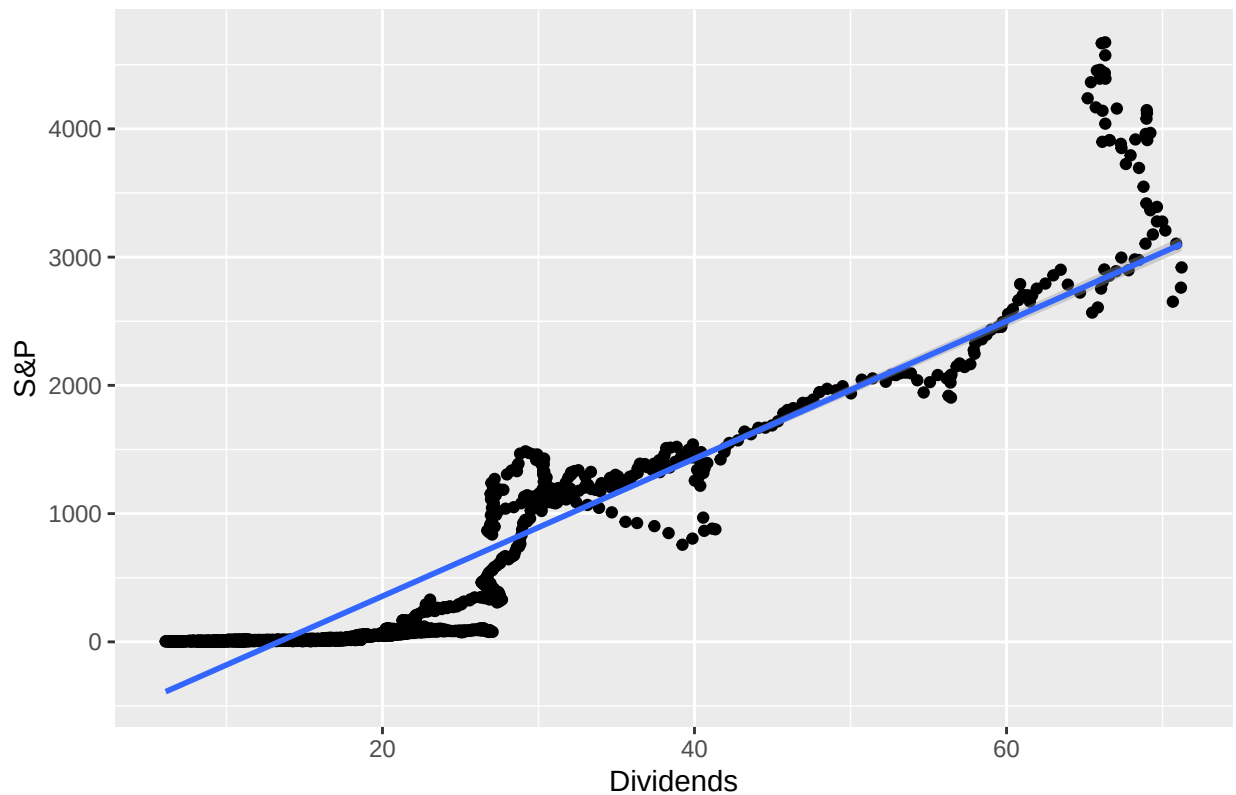


Real Earnings and the S&P are highly correlated, and the 0.94 slope of the best fit line proves this, but many things other than the S&P monthly rates go into Earnings, as can be seen by the curvy nature of the data. Although trending upwards, there are clear deviations in this graph that will provide an interesting twist to our model.

```
ggplot(data = sandp, aes(x = Div, y = SP500)) +  
  geom_point() +  
  geom_smooth(method = "lm") +  
  ggtitle("Div/S&P Scatter") +  
  labs(x = "Dividends", y = "S&P")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Div/S&P Scatter

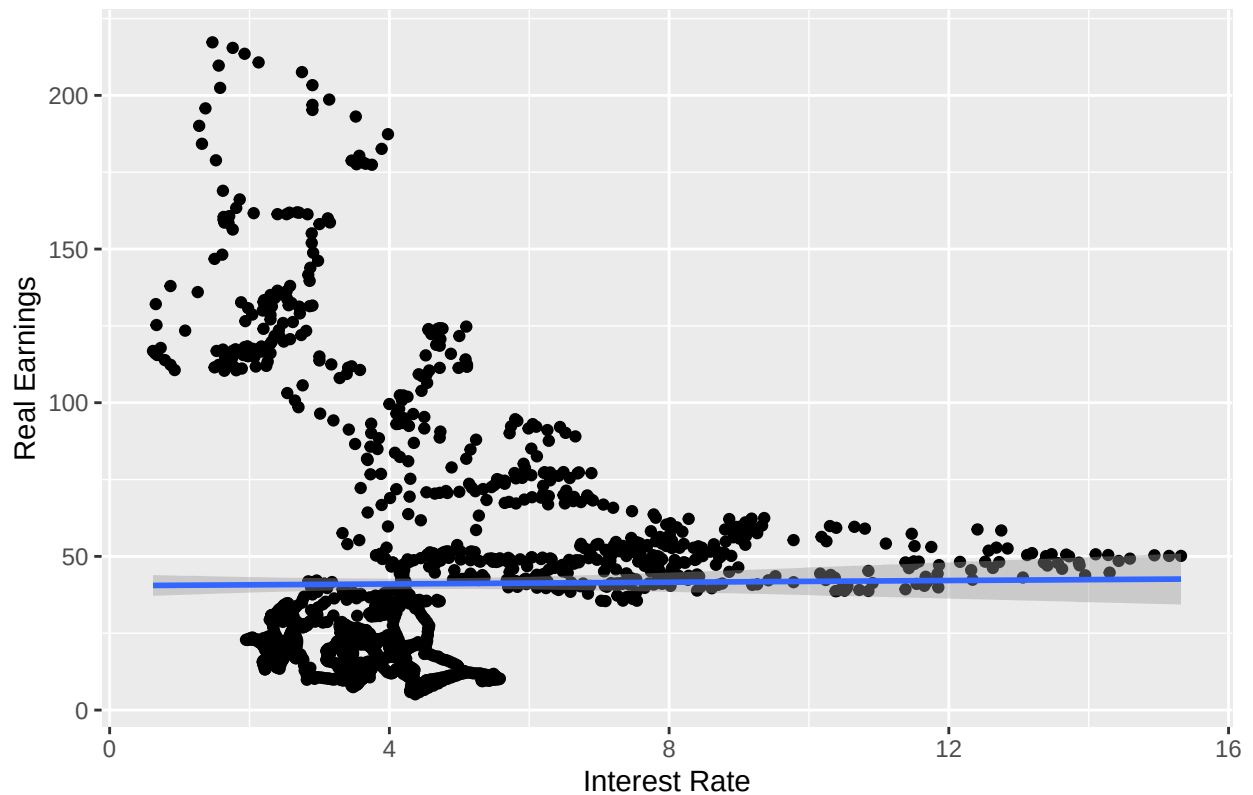


This graph features a similar pattern to the one directly above, with a sharp rise towards the end. This is because it takes many years to build the curving figure seen snaking up the line of best fit, and the S&P just recently hit 3000+ points. Dividend payments are very constant at these newer levels despite the S&P fluctuating.

```
ggplot(data = sandp, aes(x = Interest, y = Earn)) +  
  geom_point() +  
  geom_smooth(method = "lm") +  
  ggtitle("Interest Rate/Real Earnings Scatter (low alpha)") +  
  labs(x = "Interest Rate", y = "Real Earnings")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Interest Rate/Real Earnings Scatter (low alpha)

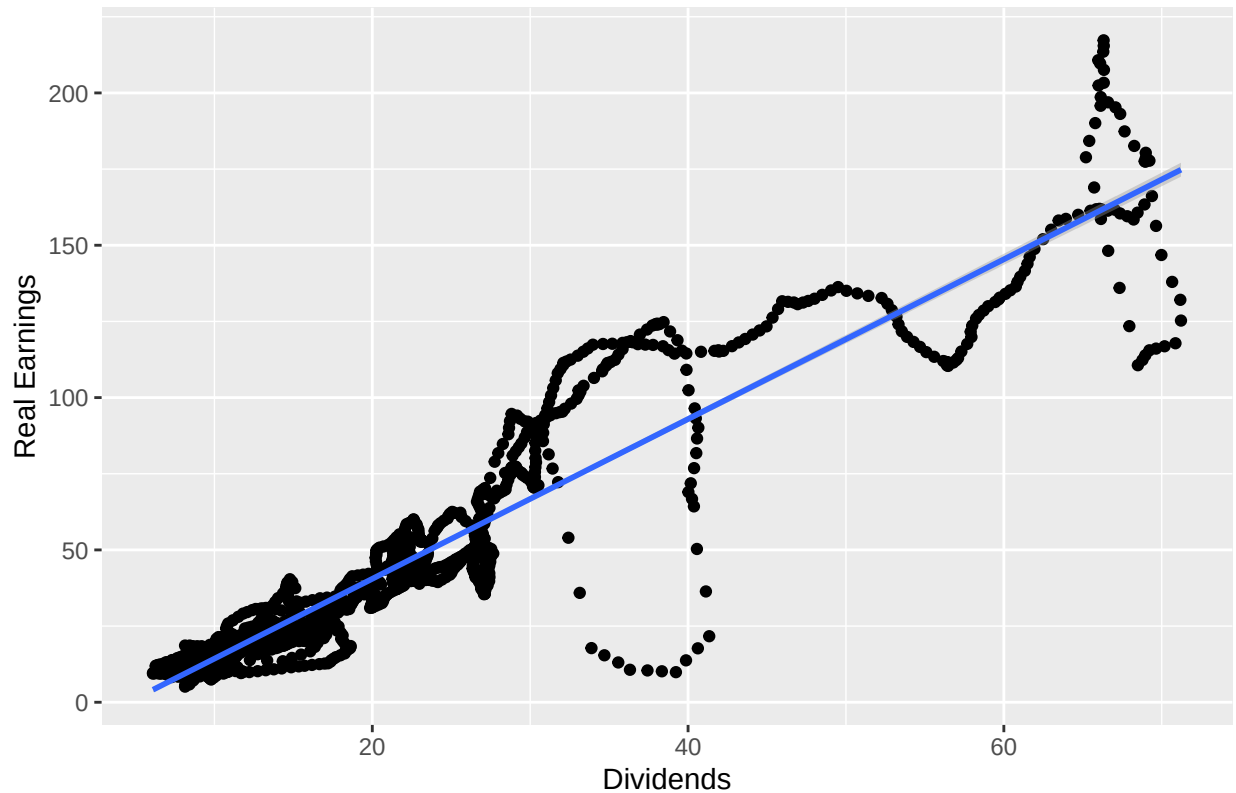


There does not seem to be a real pattern between Earnings and the Interest Rate. It seems that earnings are constant at around 50 when the Interest rate is in outlier territory, but when it settles down Earnings seem to fluctuate. My prediction is that this is because of the random walk of stock prices when interest is standard, but more analysis is needed to prove this.

```
ggplot(data = sandp, aes(x = Div, y = Earn)) +
  geom_point() +
  geom_smooth(method = "lm") +
  ggtitle("Dividends/Real Earnings Scatter (low alpha)") +
  labs(x = "Dividends", y = "Real Earnings")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Dividends/Real Earnings Scatter (low alpha)



Dividends and Earnings have a very strong, positive correlation. They don't directly predict each other, but provide useful information. Because of a stocks random walk, we see the upwards snaking pattern, but with more clarity in the more recent, higher paying years.

### 5 Number Summary

The 5 Number Summaries tend to show similar information as the Boxplots. The middle three numbers (1st Quartile, Median, 3rd Quartile), are exactly the same. The first and last numbers are the variables minimum and maximum value.

```
fivenum(sandp$SP500)
```

```
## [1] 2.730 7.930 18.020 180.900 4674.773
```

```
fivenum(sandp$CPI)
```

```
## Warning: Unknown or uninitialised column: 'CPI'.
```

```
## [1] NA NA NA NA NA
```

```
fivenum(sandp$Div)
```

```
## [1] 6.11 10.61 16.24 25.66 71.22
```



```
fivenum(sandp$Interest)
```

```
## [1] 0.62 3.15 3.80 5.10 15.32
```

**Question 2: TSDisplay**